

Ehsan Al Twal

✉ ehsantwal@gmail.com ☎ 604-657-8140 in LinkedIn ○ Github ↗ Portfolio

SKILLS

Electrical

PCB Design in Altium,
Building, Debugging,
Soldering, FPGAs, isoSPI,
UART, i2c

Programming

NN, CNN, SIFT, ROS,
CV, Java, C, C#, OOP,
HTML, Python, GUI,
Game development,
Assembly

Mechanical

Designing, Prototyping,
Testing, 3D printing, CAD
(Onshape, Solidworks)

Microprocessors

STM32F13CT,
Arduino, De1-SoC

EDUCATION

3rd Year UBC Engineering Physics Undergrad, University of British Columbia

Sep 2021 – Jun 2026

- Engineering Physics is the pursuit of taking new ideas from concept to practical use, that also combines physics, mathematics and engineering.
- Key Courses include: software construction; mechanics; electronics; and design

Vancouver, Canada

PROFESSIONAL EXPERIENCE

Mechanical/Materials Engineer, Terrella Energy Systems - Matthews International

Jan 2023 – Apr 2023

- Lead a project to optimize the behavior of a material by measuring/testing its mechanical properties and studying its behavior under different environments.
- Developed new mechanical testing methods and techniques, identified equipment needed for the test, and conducted the experiments which resulted in useful findings
- Fixed a vacuum chamber using improvised materials.

Vancouver, Canada

Mechanical Engineering Trainee, Fine Hygienic Holding

Jul 2022 – Aug 2022

- Demonstrated high safety awareness while successfully measuring vibration values for predictive maintenance testing of heavy machinery.
- Trained in a Paper Mill factory for maintenance with exposure to digital control systems, production lines, and manufacturing process reports.

Maintenance/ Waiter/ Life guard, Mariam Hotel

2017 – 2022

- Experienced in general maintenance, food service, lifeguarding, and social media account management.
- Strong customer service skills and attention to detail.

STUDENT DESIGN TEAMS

UBC Formula Electric, Altium Designer, PCB Design, LTSpice, Jira

- Designed the BMS Monolithic Module PCBs using Altium Designer for balancing and monitoring the voltage and temperature of the battery pack, which encompassed PCB design, assembly, component selection, and simulation via LTSpice.
- Engineered a test setup to simulate li-ion battery packs for precise evaluation of the BMS Monolithic Module and executed effective PCB debugging to ensure optimal functionality.
- Performed online verification and validation of various PCB schematics through Jira and developed a prototype of the accumulator enclosure, specifically aimed at facilitating harnessing activities.

UBC Open Robotics, OnShape, Solidworks

- Mechanical Team Lead for the Gripper Project.
- Effectively delegated responsibilities and managed project workflow.
- Spearheaded the design and prototype development of the claw's motion system.
- Demonstrated expertise in CAD design, utilizing OnShape for project assignments.

PROJECTS

Robot Race Car Competition, STM32, C++, Circuit Design & Debugging, OnShape CAD, Sensor Fusion, PID

- Built the CAD and fabricated a sturdy chassis using sheet metal techniques.
- Pioneered an Acronym Steering System for enhanced track performance and agility.
- Engineered an advanced H Bridge circuit for precise motor control and maneuvering; extensively tested, soldered, and noise issues expertly resolved.
- Integrated essential electronic circuits, including a rotary encoder, servo motor, and regulators.
- Fused sensor arrays (magnetometer, IMU, accelerometer, gyroscope) for 3D orientation determination.
- Utilized laser distance and ultrasound sensors to pinpoint track positions, enabling accurate localization.
- Devised a strategic track mapping approach via Desmos, programming the robot for autonomous route navigation.

Autonomous ROS Robot with CNN-Based Text Recognition, Python, CNN, Computer Vision, ROS, PID, Linux

- Programmed a vehicle to navigate autonomously while obeying traffic laws in a ROS environment.
- Implemented OpenCV for real-time detection of pedestrians, cars, road elements, and roadside posters featuring embedded clues.
- Developed a convolution neural network to read and interpret text from images, essential for clue analysis.

Mobile Game, C#, Unity Engine

- Self-taught game development, programming, and graphics production through online videos and courses.
- Conceived and developed a mobile game idea independently.
- Demonstrated proficiency in game development, using Unity for game engine, C# for programming, and Blender for graphics and assets.

Robot Claw, Arduino, C++

- Designed, Modeled, Prototyped (C-sketch & Physical), and Built two robot claws using cardboard, tape, servo motor, Arduino, IR sensor, and a Joystick
- Coded in C language targeting Arduino platform to control Motion, Measure Distance, and Gripping aspects
- Scored the highest in class competition

Robot Tank (Personal), Arduino, C++, Bluetooth

- Built and wired robotic tank involving various components such as servo motor, Ultrasonic sensor, Bluetooth module, LED lights, and more
- Developed code in C targeting the Arduino platform to control Motion, Tracking, and Lighting aspects
- Overcame faults in the product by carving wheels

AWARDS

Dean's Honour List, University of British Columbia

Trek Excellence Scholarship, University of British Columbia

100% High School Average

graduated highschool with an average of 100%

Aug 2021

14 Medals in Track & Field and Soccer